Quantification and Classification under Dataset Shift

Proposal for a Workshop at ECML-PKDD 2026

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Abstract. *Quantification* and *classification* are two supervised learning tasks that are both hampered by changes in data distributions, so-called *dataset shifts*. The two tasks differ in what they learn to predict: while classification predicts the class label of each individual data point, quantification deals with the prediction of the prevalence (i.e., the relative frequency) of each class in every unlabelled set of data. Predicting this prevalence is worthwhile whenever it can change from set to set, due to various types of dataset shift that may exist between these sets and the training data. Both classifiers and quantifiers can suffer from these shifts, at least as long as they are not designed to handle the current type of shift robustly. Notably, a robust quantifier can facilitate robust classification and a robust classifier can facilitate robust quantification under dataset shift.

Having organized five editions of the workshop "LQ: Learning to Quantify", we are now aiming for a workshop with a broader scope, which does not only include quantification under dataset shift but also classification under dataset shift. This choice came naturally, having observed recent trends in quantifiers being used to mitigate shift in classification and vice versa. Due to the immense potential for further cross-fertilization, we consider the joint discussion of both topics—within a novel workshop format—as the ideal way forward.

(Important) Premise: This is a proposal for a full-day workshop. However, should the organisers feel that, due to a shortage of rooms or due to other reasons, a full day cannot be allocated, we would accept our proposal to be downsized to a half-day event.

1 Quantification & Classification under Shift

A traditional, despite often over-simplifying assumption in supervised learning is that all data, during training and during model deployment, are independently drawn from the same data distribution. This IID (independent and identically distributed) assumption is only valid in highly controlled settings that practical applications often lack. As a consequence, applying a "standard" learning algorithm (one that assumes IID data) to data that actually is non-IID can result in severe failures of the learned models. Suppose, for instance, a classifier is trained with more positives than negatives; applying it to a context with *less* positives than negatives, we can expect performance degradation, unless we properly adapt the classifier to this type of dataset shift.

The proper handling of non-IID data, particularly those where training and deployment scenarios differ, has been investigated in learning under dataset shift. This learning task is often also called *domain adaptation* and *out-of-distribution learning*. In this workshop, we are interested in the classification tasks from this realm, due to their close connection to quantification and due to the potential benefits that stem from further exploring this connection.

1.1 Quantification under Shift

In a number of applications involving class membership of data points, the final goal is not to determine to which class (or to which classes) the individual, unlabelled instances belong, but to estimate the prevalence (or "relative frequency", or even "prior probability") of each class in unlabelled sets of data. Estimating class prevalence for unlabelled data via supervised learning is known as *quantification* (or "learning to quantify", or "class prior estimation").

Quantification has several applications in fields that are inherently interested in characterising *aggregations* of individuals, rather than the individuals themselves. Such fields include the social sciences, market research, epidemiology, and others. As we might put it, disciplines like these are usually not interested in finding the needle in the haystack, but in characterising the haystack itself. For instance, most applications of tweet sentiment classification are not concerned with estimating the true sentiment of individual tweets; rather, they are concerned with the relative frequencies of these classes (which might shift) in the set of unlabelled tweets under study.

It has been unequivocally shown that performing quantification by simply counting the class predictions for individual instances (the "classify and count" method), may lead to poor quantification accuracy. This is due to the presence

of dataset shift, but also due to a mismatch between the loss functions of quantification and classification. As a result, quantification has evolved as a task of its own, different from classification in its goals, methods, techniques, evaluation measures, and evaluation protocols (Esuli et al., 2023; González et al., 2017). While quantification can be solved effectively even without classifying individual instances at all, today's best-performing methods employ classification as a pre-processing step, before solving the actual quantification task.

1.2 Classification under Shift

In classification, the goal is to predict the class membership of any individual instance. Under dataset shift, the performance of these predictions can degrade. Most studies on dataset shift are concerned with such predictions for individual instances (i.e., with classification and regression) rather than quantification.

Several types of dataset shift have been characterized (Kull and Flach, 2014), each with their own effects on supervised learning. Some of these types allow for classifier adaptations that mitigate these effects and, hence, facilitate accurate classification even despite dataset shift. Shift types that have been shown to allow for being mitigated include covariate shift (addressing changes in $\mathbb{P}(X)$; see Sugiyama et al., 2007), specific sub-types of concept shift (addressing changes in $\mathbb{P}(Y \mid X)$ under additional assumptions; see, e.g., Bouvier et al., 2019), and label shift (addressing changes in $\mathbb{P}(Y)$; see Saerens et al., 2002).

One might argue that quantification would be solved with a perfectly shift-robust classifier. However, in the case of label shift, quantification is in fact a prerequisite for shift-robust classification (Donyavi et al., 2025; Saerens et al., 2002): as soon as we have accurately predicted the shifted class prevalences of the target data (as it is done in quantification), and be this prediction with or without classification as a pre-processing step, we can easily adapt any classifier to the new class prevalences and, hence, to the label-shifted data. This is because dataset shift is an inherent issue of data distributions and because quantification inherently deals with such distributions, rather than individual data points. Hence, we expect further advances in classification under various types of shift to be facilitated by applying quantification. Beyond this expectation, our workshop also welcomes all other submissions on classification under shift.

2 The 1st International Workshop on Quantification and Classification under Dataset Shift

The goal of this **full-day workshop** is to bring together all researchers interested in methods, algorithms, evaluation measures, and evaluation protocols for quantification and classification under dataset shift (QCDS), as well as practitioners who face the challenges of dataset shift in their applications.

The main topics on which contributions will be solicited, and which will form the main themes of the workshop, will be:

- Binary, multiclass, multilabel, and ordinal QCDS
- Semi-supervised / transductive QCDS
- Multi-instance learning / learning for set-structured data
- Deep learning for QCDS
- Characterizations of dataset shift, and their impact on QCDS
- Detecting and measuring (different types of) dataset shift
- Evaluation measures and protocols for QCDS
- Improving classification under dataset shift via quantification
- Classifier calibration under dataset shift
- New datasets and applications for evaluating QCDS

Discussing both quantification and classification under dataset shift in a single workshop has a huge potential for cross-fertilization between the two topics. This potential is verified by methods that use quantification for designing shift-robust classifiers (Donyavi et al., 2025) and by methods that jointly train a robust quantifier and a robust classifier (Saerens et al., 2002). Since quantification inherently deals with distributions of data, and dataset shift describes the behaviour of distributions, we expect further advances at the very intersection of quantification and classification under shift.

2.1 Previous events

This workshop is the successor of the International Workshop on Learning to Quantify (LQ), which we organized in five editions. The 1st edition was co-located with the 30th ACM Conference on Information and Knowledge Management (CIKM 2021), while the 2nd–5th editions have been co-located with ECML-PKDD 2022–2025. More information on these editions can be found online and in the workshop proceedings:

- 1st edition: `https://cikmlq2021.github.io/` (del Coz et al., 2021)
- 2nd edition: `https://lq-2022.github.io/` (del Coz et al., 2022a)
- 3rd edition: `https://lq-2023.github.io/` (Bunse et al., 2023a)
- 4th edition: `https://lq-2024.github.io/` (Bunse et al., 2024)
- 5th edition: `https://lq-2025.github.io/` (Bunse et al., 2025)

These previous editions had between 20 and 40 participants each. They were organized as half-day workshops, with most of them being supplemented by a half-day tutorial on LQ.

Related initiatives, which witness the significant interest in quantification, are the 1st and 2nd editions of the CLEF "Learning to Quantify" lab (LeQua 2022 and LeQua 2024; see `https://lequa2022.github.io/` and `https://lequa2024.github.io/`), a data challenge explicitly devoted to quantification (Esuli et al., 2022, 2024). Note also that the ECML-PKDD 2023 Best Student Paper Award was given to a paper on quantification (Dussap et al., 2023).

The new workshop on QCDS broadens the quantification focus of LQ to the more general theme of classification under dataset shift. We deeply believe that

this new breadth will be even more attractive for the ECML-PKDD community, since (a) its main topics squarely fall within the machine learning and data mining areas, (b) quantification is still a relatively unknown task that many ECML-PKDD attendees can profit from, and (c) dataset shift is a highly relevant and broad topic that holds fruitful interactions with other areas of ML and DM.

2.2 Timeline and Call for Papers

The timeline of this workshop is identical to the general timeline of workshops at ECML-PKDD.

Call for Papers published & advertised:	March 10, 2026
Paper submission deadline:	June 5, 2026
A/R notification deadline:	June 29, 2026
Final copy submission deadline:	July 10, 2026
Workshop:	September 7 or 11, 2026

The Call for Papers for this workshop will mirror the contents of this workshop proposal.

2.3 Workshop structure

We plan the full-day workshop to consist of:

1. A keynote talk by a renowned and thought-provoking scientist. Our current working hypothesis is to invite Prof. Amos Storkey (University of Edinburgh, UK), a leading machine learning expert who has given important contributions to research on dataset shift that are also highly relevant to quantification.
2. Presentations of submitted papers. We plan to allow 20 minutes per paper.
3. A final session of open discussion on the main themes that have emerged during the previous talks and in the ensuing Q&A sessions, and on the directions that the QCDS workshop and the LeQua data challenge should take.

2.4 Program Committee

We plan to assemble a small program committee, whose task will consist of reviewing the submitted papers and advising on other decisions. Our current hypothesis is to invite in the PC the following people (names in **boldface** have already agreed to serve on the PC, should the workshop proposal be accepted – the others have not answered yet):

1. Amr Alexandari, Stanford University, US
2. Haoyue Bai, University of Wisconsin-Madison, US
3. **Gustavo Batista, University of New South Wales, AU**
4. Pawel Piotr Czyz, ETH AI Center, CH

5. **Clemens Damke, Ludwig Maximilian University of Munich, DE**
6. **Juan José del Coz, University of Oviedo, ES**
7. Zahra Donyavi, University of New South Wales, AU
8. **Andrea Esuli, Consiglio Nazionale delle Ricerche, IT**
9. **Cèsar Ferri, Universitat Politècnica de València, ES**
10. **Peter Flach, University of Bristol, UK**
11. Wei Gao, Singapore Management University, SG
12. Saurabh Garg, Thinking Machines, US
13. Devin Guillory, UC Berkeley, US
14. **Barbara Hammer, University of Bielefeld, DE**
15. **Rafael Izbicki, Federal University of São Carlos, BR**
16. **Mira Jürgens, Gent University, BE**
17. Zachary Lipton, Carnegie Mellon University, US
18. **André Maletzke, Universidade Estadual do Oeste do Paraná, BR**
19. **Olaya Pérez-Mon, University of Oviedo, ES**
20. **Teodora Popordanovska, KU Leuven, BE**
21. **Marco Saerens, Catholic University of Louvain, BE**
22. **Tobias Schumacher, University of Mannheim, DE**
23. Amos Storkey, University of Edinburgh, UK
24. **Dirk Tasche, North-West University, SA**
25. **Lorenzo Volpi, Consiglio Nazionale delle Ricerche, IT**
26. **Willem Waegeman, Ghent University, BE**

This committee is a varied set, since it includes people with different backgrounds (machine learning, data mining, information retrieval, statistics); all of these people have given substantial contributions to the dataset shift and quantification literature. Some of them have already been available as reviewers during former editions of the LQ workshop and some of them have been newly acquired for the classification-related part of QCDS.

Submissions will be reviewed in single-blind mode, i.e., authors will not be required to hide their identity, but will be allowed to do so if they wish. Submitted papers will go through a standard review process, with each paper being reviewed by three PC members. The final acceptance/rejection decisions will be taken by the organisers based on the three reviews received from the PC.

2.5 Proceedings

The proceedings of LQ for the editions 2022–2025 have been self-published, i.e., they are available for free download from the LQ websites. We intend to do the same for this workshop, in order to avoid putting them behind a paywall and to leave them open-access.

If, however, the joint workshop proceedings with Springer is a requirement for workshops at ECML-PKDD, we are also happy about taking this new opportunity.

3 The Organising Committee

Mirko Bunse obtained his PhD in Computer Science in 2022 and is now holding a PostDoc position at TU Dortmund University, Germany. His research focuses on quantification, active learning, and label noise, with applications in astroparticle physics. Since 2022, Mirko is one of the directors of the Working Group on Physics, Modern IT and Artificial Intelligence within the German Physical Society. He was a co-organiser of the 3rd, 4th, and 5th editions of the International Workshop on Learning to Quantify (LQ 2023–2025) (Bunse et al., 2023a, 2024, 2025), of the 7th, 8th, and 9th editions of the International Workshop on Interactive Adaptive Learning (IAL 2023–2025), and of the Workshop on Machine Learning for Astroparticle Physics and Astronomy at the German INFORMATIK conference in 2022.

Pablo González obtained his PhD in Computer Science in 2019 and is currently an Assistant Professor at the University of Oviedo (Spain). His main research interests include deep learning and quantification and the application of these techniques to plankton data analysis. He was a co-author of the first survey paper on learning to quantify (González et al., 2017), and a co-organiser of all five editions of the International Workshop on Learning to Quantify (LQ 2021–2025) (Bunse et al., 2023a, 2024, 2025; del Coz et al., 2021, 2022a).

Eyke Hüllermeier is a full professor at the Institute of Informatics at LMU Munich (Germany), where he heads the Chair of Artificial Intelligence and Machine Learning. Before joining LMU, he held professorships at the universities of Dortmund, Magdeburg, Marburg, and Paderborn, and spent two years as a Marie Curie Fellow at the IRIT in Toulouse, France. He received his Habilitation degree from Paderborn University in 2002, and his PhD in 1997. His research interests center on methods and theoretical foundations of machine learning and reasoning under uncertainty. He has published over 300 articles on these topics in leading journals and at major international conferences, where several of his contributions have been recognized with scientific awards. Prof. Hüllermeier is President of the European Association for Data Science (EuADS) and Editor-in-Chief of Data Mining and Knowledge Discovery, one of the leading journals in the field of AI. He is also an associate editor of the IEEE Transactions on Pattern Analysis and Machine Intelligence, and serves on the editorial board of several other journals, including Machine Learning and the Journal of Machine Learning Research. He is a fellow of IFSA and ELLIS and affiliated with the German Research Centre for Artificial Intelligence (DFKI, Kaiserslautern).

Alejandro Moreo is a tenured Senior Researcher at the Institute for the Science and Technologies of Information of the Italian National Council of Research (ISTI-CNR). He has obtained a PhD in Computer Science and Information Technologies from the University of Granada, Spain, in 2013. His research interests lie at the interface of data mining and machine learning, with particular emphasis on deep learning, representation learning, and transfer learning for cross-lingual classification. He is a member of the Editorial Boards of the Journal of Machine

Learning Research, the ACM Journal on Computing and Cultural Heritage, and PeerJ Computer Science. He has taught tutorials on learning to quantify at SIGIR 2019, FQAS 2019, SocInfo 2020, ECML-PKDD 2023 and ECML-PKDD 2024. He is a co-author of the recent book "Learning to Quantify" (Springer Nature) (Esuli et al., 2023), a co-organiser of the LeQua 2022 and LeQua 2024 shared tasks on Learning to Quantify (Esuli et al., 2022, 2024), and a co-organiser of all five editions of the International Workshop on Learning to Quantify (LQ 2021–2025) (Bunse et al., 2023a, 2024, 2025; del Coz et al., 2021, 2022a).

Fabrizio Sebastiani is a tenured Director of Research at the Institute for the Science and Technologies of Information of the Italian National Council of Research (ISTI-CNR), a former Principal Scientist at the Qatar Computing Research Institute, and a former Associate Professor at the University of Padova, Italy. His research interests lie at the intersection of machine learning, information retrieval, and natural language processing, with particular emphasis on quantification, text classification, authorship analysis, and technology-assisted review. He has been a General Chair or co-Chair of CLEF 2022, ECIR 2021, SIGIR 2016, SPIRE 2011, and ECIR 2003; a Program Chair or co-Chair of AFIRM 2020, ECDL 2010, SIGIR 2008, and ECIR 2003; and the Vice-Chair of ACM SIGIR from 2003 to 2007. He has been a co-Editor-in-Chief and Founding Editor of Foundations and Trends in Information Retrieval (Now Publishers). He is a Senior Associate Editor of the ACM Transactions on Information Systems (ACM Press), and a former Associate Editor / member of the Editorial Board of about 15 other journals. He has taught tutorials on learning to quantify at SIGIR 2019, FQAS 2019, SocInfo 2020, and ECML-PKDD 2023. He is a co-author of the recent book "Learning to Quantify" (Springer Nature) (Esuli et al., 2023), a co-organiser of the LeQua 2022 and 2024 shared tasks on Learning to Quantify (Esuli et al., 2022, 2024), and a co-organiser of all five editions of the International Workshops on Learning to Quantify (LQ 2021–2025) (Bunse et al., 2023a, 2024, 2025; del Coz et al., 2021, 2022a).

4 Other Relevant Information

We plan to disseminate the tutorial call for participation and workshop call for papers

- through the social media channels of the tutors/organisers;
- through the main mailing lists of the scientific areas involved (ML, DM, IR, NLP);
- through a dedicated mailing list of researchers who have contributed to the quantification literature, that we have incrementally assembled when publicising the five editions of the LQ workshop (LQ 2021–2025).

After the workshop has taken place, we plan to write—similarly to (Bunse et al., 2023b; del Coz et al., 2022b)—a workshop report to be submitted for publication at SIGKDD Explorations.

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